



Swin transformer with attention mechanism: a novel framework for person re-identification

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Abstract

Vision Transformers have successfully captured long-range dependencies across various computer vision tasks, including Person Re-Identification (PReID). This paper proposed SwinReID, a novel PReID framework based on the Swin Transformer (ST) architecture. Unlike traditional convolutional neural networks (CNNs) that process entire images, SwinReID divides images into small, non-overlapping patches. This patch-based strategy enables a comprehensive understanding of global image context while preserving fine-grained spatial details, addressing challenges such as occlusions and appearance variations in PReID tasks. Conventional CNNs struggle with long-range dependencies due to their localized receptive fields and downsampling operations, limiting their effectiveness in complex scenarios. SwinReID overcomes these limitations by integrating an advanced transformer-based backbone and a lightweight channel attention (CA) module, which selectively improves discriminative feature channels. This refinement leads to better feature representation and robustness, particularly in challenging environments. SwinReID achieves state-of-the-art performance in benchmark datasets, with rank-1 accuracies of 95.5% in Market1501, 89.9% in DukeMTMC-reID, 64.1% in Occluded-Duke, and 91.63% in MSMT17, demonstrating its effectiveness in various and challenging PReID scenarios.

Keywords Person re-identification (P-ReID) · Resolution disparity issue · Lower-resolution P-ReID (LR-P-ReID) · Image enhancement

1 Introduction

Modern surveillance systems improve public safety by supporting tasks like crime prevention, identifying individuals, locating missing persons through advanced cameras,

and generating large amounts of video datasets. Manually processing these datasets to identify individuals is impractical, creating the need for automated solutions like PReID. PReID aims to match individuals across non-overlapping camera views (see Fig. 1), which is critical for applications such as multi-camera tracking and behavior analysis [1]. The primary challenge in PReID is to extract robust features that account for variations in pose, illumination, occlusion, and resolution.

Most traditional PReID techniques, which relied on handcrafted feature extraction [2, 3] and similarity matching measurement [4, 5], were inefficient, time-taking, error-prone, and struggled with generalization. Deep learning-based methods such as CNN, utilizing architectures like ResNet [6] and DenseNet [7], VGGNet [8], and GoogleNet [9] improved the PReID's performance with the help of extracting local features by loss functions including triplet loss [10]. Despite notable advances in deep learning, CNNs are inherently constrained by their local receptive fields, limiting their ability to capture long-range dependencies and handle significant variations in appearance, such

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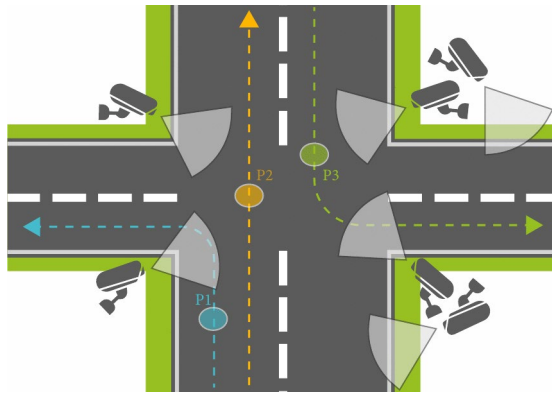


Fig. 1 Non-overlapping cameras

as pose changes, illumination, and occlusions. As a result, CNN-based methods often struggle to maintain robustness in complex Person Re-Identification (PReID) scenarios. In contrast, transformer-based models, such as the Swin Transformer [11], leverage self-attention mechanisms to capture both global and local dependencies seamlessly. This enables them to model long-range relationships more intuitively, providing improved performance and robustness over traditional CNN-based methods, particularly in challenging and dynamic environments. Motivated by the recent success of Transformer architectures, particularly the Swin Transformer (ST), and recognizing the limitations of CNNs in capturing global dependencies for complex PReID scenarios, we proposed SwinReID, a novel PReID framework. SwinReID integrated the ST with a lightweight channel attention mechanism inspired by the Squeeze-and-Excitation (SE) module. This integration strengthens feature representation by capturing both fine-grained details and global contextual information, leading to boosted robustness and adaptability in challenging ReID scenarios. The novel SwinReID network introduced also employed optimization techniques such as Random Erasing (RE), linear Warm-Up (LW-up), and three different losses to optimize performance. The SwinReID approach achieves state-of-the-art Rank-1 accuracies of 89.9% on DukeMTMC-ReID, 91.63% on MSMT17, and 95.5% on Market-1501. The key contributions of this study are as follows:

1) We proposed **SwinReID**, a transformer-based framework with lightweight channel attention to overcome CNN limitations, improving fine-grained feature capture and global context for PReID.

2) A novel channel attention mechanism is introduced, capturing discriminative features to improve the ST model's ability to handle occlusions and appearance variations, increasing robustness in challenging scenarios.

3) **SwinReID** achieved state-of-the-art performance on MSMT17, Market1501, DukeMTMC-reID, and Occluded-Duke, proving its effectiveness in real-world surveillance.

2 Related work

2.1 CNN-driven person re-identification (PReID)

In the area of object detection and PReID, CNNs have become essential due to their strong feature extraction capabilities for retrieval tasks. CNN-based techniques usually learn global or local features and then combine them for efficient retrieval. While Zheng et al. [12] tackled PReID as a multi-class classification problem, which became the standard approach, PersonNet [13], and other early methods used small convolution kernels to capture fine-grained details. Multiscale feature representations were introduced by Qian et al. [14] to improve feature diversity, substantially improving retrieval performance. Wang, Guanshuo, et al. [15] introduced Multiple Granularity Networks (MGN) in Part-based Convolutional Baseline [16], and AlignedReID++ [17] segmented images to extract localized features, providing a granular understanding of individual components.

Furthermore, recent CNN-based PReID methods address occlusions, clothing changes, and cross-modality challenges. OC4-ReID [18] introduces new datasets and a Part-Robust Triplet loss for occluded cloth-changing ReID. Semantically Enhanced ReID [19] employs human parsing and spatial-frequency interaction for occlusions, while Spectral Enhancement [20] improves infrared-visible ReID. Attention-Enhanced Disentanglement [21] separates identity from clothing features using attention mechanisms. These advances considerably strengthen CNN-based ReID in real-world scenarios.

Similarly, the increasing integration of attention mechanisms has considerably improved the problem of re-identification by improving feature discrimination. Some challenges in PReID, such as pose variations and detection errors, have been effectively addressed in PReID by incorporating attention mechanisms. Zhang et al. [22] proposed global attention to address the impact of useless background noise. Chen et al. [23] further enhanced the attention mechanisms to increase feature discriminability and mitigate the limitations in zero-shot learning situations. These advancements reflect the evolution of PReID methodologies, combining CNNs and attention mechanisms to achieve state-of-the-art performance in complex retrieval tasks.

2.2 Occlusion person re-identification (PReID)

PReID in real-world situations is challenging, particularly when a person is occluded. Several methods aim to recover missing information and adjust visible parts to improve identification. Pose assessment and body segmentation [24] are commonly used to strengthen occluded body regions,

while Generative Adversarial Networks (GANs) [25] simulate and fill in missing regions. Gao et al. [24] introduced Pose-guided Visible Part Matching (PVPM) to focus on discriminatory local features for better re-identification. Zhuo et al. [26] contributed key datasets and attention-based networks for occlusion handling. Miao et al. [27] developed the Occluded-Duke dataset and a pose-guided feature alignment technique to combine global and local features for augmented robustness.

Similarly, in medical imaging, lightweight models like RL-Cervix.Net [28] demonstrate the potential of reinforcement learning for handling occlusions and variations in specialized datasets, offering valuable insights for future PReID work.

2.3 Vision transformers and attention mechanisms

Transformers are initially designed for natural language processing [29]. They have shown strong potential and are successfully utilized in computer vision problems such as object detection and image classification. Their self-attention mechanism captures long-range dependencies and global context. This makes them suitable for PReID problems, which acquire global and fine-grained features.

The Vision Transformer (ViT) [30] was a pioneering model to process images by treating them as patch sequences, capturing global and local information. The DeiT network [31] was further enriched by ViT to introduce a method distillation approach, enabling them to act well even with limited datasets. These algorithms produce global and local image tokens to improve feature extraction for re-identification problems. To better integrate local features, the Locally Aware Transformer (LA-Transformer) [32] incorporates ViTs with convolutional networks such as Parts-based Convolution Baseline (PCB) [16]. This hybrid technique boosts the network's power to focus on both global and local features. The TransReID network [33] is further empowered with transformer-based PReID by directly employing local and global representations, achieving state-of-the-art performance.

In addition to SENet [34], various attention mechanisms have been adopted in vision tasks. CBAM [35] integrates channel and spatial attention, while Non-Local Attention [36] captures long-range dependencies. Zhang et al. [22] proposed RGA for strengthened feature learning. Among these approaches, the SENet is utilized in our proposed model for its simplicity and effectiveness in adaptively recalibrating channel-wise feature importance, making it particularly well-suited for ReID tasks that require precise discriminative feature learning.

The ST [11], a hierarchical transformer, leverages shifted windows for self-attention to improve execution

performance while it's efficient in capturing both local and global contexts. The ESwin-Transformer [37] combines convolutional modules, illustrating specifically efficient for smaller datasets. Moreover, a channel attention mechanism technique has proven crucial for enhancing transformer performance by prioritizing informative channels for feature extraction to improve the channel-wise features.

Furthermore, hybrid techniques such as mixed local-channel attention, combining spatial and channel attention, have illustrated improved performance in object-detection problems [38]. These advanced developments in attention mechanisms, mainly focusing on channel-wise information, are highly relevant to the PReID problem.

We proposed SwinReID, a network that leverages the ST [11] backbone and incorporates a channel-wise attention mechanism [34] to prioritize discriminative channels for refined feature extraction. This approach improves PReID performance by capturing both fine-grained information and global context across datasets with varying resolutions. The fusion of CNNs, attention mechanisms, and transformers has advanced PReID, enhancing robustness and accuracy.

3 Methodology

This section describes the SwinReID Channel Attention Model for PReID. Section 3.1 presents the architecture, combining the Swin Transformer and SE-Block Channel Attention with techniques like RE, LW-up, circle (CL), ArcFace (AFL), CosFace (CFL), and Contrastive (CTL). Section 3.2 details the Swin Transformer (ST) Block, Section 3.3 covers the SE-Block Channel Attention, and Section 3.4 discusses optimization losses, including ArcFace Loss, CosFace Loss, and Contrastive Loss.

3.1 Proposed SwinReID architecture

We proposed a novel SwinReID framework for PReID that integrates an ST backbone with SE blocks for amplified feature representation. This design combines the ST's self-attention mechanism with SE blocks' channel recalibration to address PReID challenges in feature extraction and identification. The proposed SwinReID architecture is shown in Fig. 2. (i) a ST for hierarchical feature extraction, (ii) SE for channel-wise-attention, and (iii) three key techniques RE, LW-up, and CL to improve generalization and robustness, optimizing the PReID network. It progressively refines features across scales, emphasizing discriminative elements for robust PReID performance. Below, we detail the components and processes of our architecture.

First, the image of an input person $I \in \mathbb{R}^{H \times W \times 3}$ is divided into non-overlapping patches P_i of size $4 \times 4 \times 3$

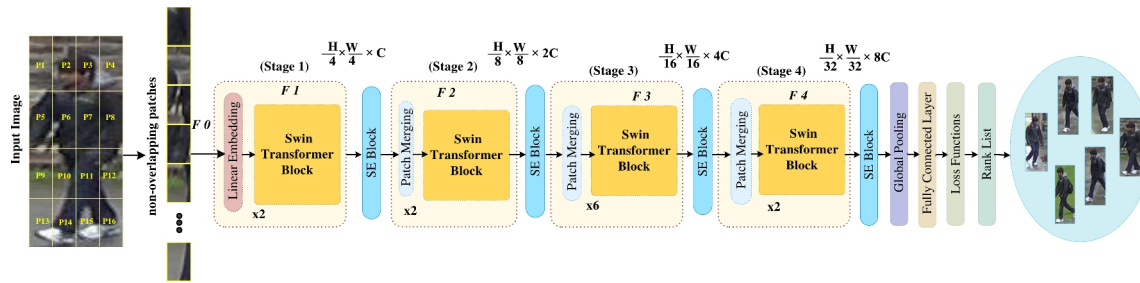


Fig. 2 The demonstration of the proposed SwinReID network's architecture

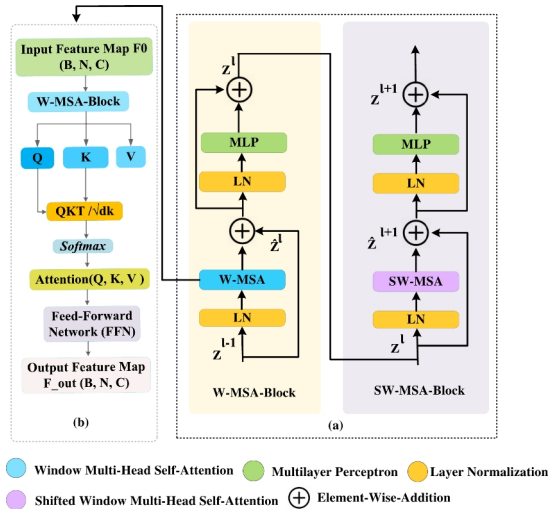


Fig. 3 Swin transformer blocks

utilized a Patch Partition component. Every single patch P_i is treated as a token and linearly embedded into a feature vector of dimension C , forming the initial representation $F_0 \in \mathbb{R}^{\frac{H}{4} \times \frac{W}{4} \times C}$. This representation is the hierarchical ST feature extraction input.

At **Stage-1**, F_0 is purified through two ST blocks. Each block utilizes the shifted-window-based multi-head self-attention (W-MSA) mechanism, which computes relationships between tokens using query (Q), key (K), and value (V) matrices, as illustrated in Fig. 3(b). The self-attention mechanism is expressed as:

$$\text{Attention}(Q, K, V) = \text{Softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V,$$

Where d_k denotes the dimension of the key vectors. Subsequent to the attention computation, feedforward networks apply linear transformations and activation functions for feature improvements.

After all ST blocks, a SE block is incorporated to recalibrate the channel-wise feature responses by employing a SE operation. The SE operation is defined as:

$$z = \sigma(W_2 \cdot \text{ReLU}(W_1 \cdot \text{GlobalAvgPool}(F))), \quad (1)$$

where W_1 and W_2 are weight matrices, σ is the sigmoid activation, and F is the input feature map. The recalibrated features form $F_1 \in \mathbb{R}^{\frac{H}{4} \times \frac{W}{4} \times C}$.

At **Stage-2**, a patch merging layer downsamples F_1 to size $\frac{H}{8} \times \frac{W}{8} \times 2C$, producing F_2 . This step reduces spatial dimensions while doubling the number of channels to retain information. F_2 passes through two ST blocks, and after these blocks, an SE block is applied for boosted channel recalibration.

In **Stage-3**, the patch merging layer further downsamples F_2 to $\frac{H}{16} \times \frac{W}{16} \times 4C$, yielding F_3 . To capture fine-grained features, it F_3 is processed through six ST blocks. After all the ST blocks, an SE block is applied to focus on important channel-wise features, ensuring critical information is retained.

In **Stage 4**, patch merging reduces the feature map to $\frac{H}{32} \times \frac{W}{32} \times 8C$, producing F_4 . This final stage includes two ST blocks, followed by an SE block, which optimizes the features for robust representation.

Finally, the feature map F_4 undergoes global average pooling to produce a compact global feature vector $G \in \mathbb{R}^{8C}$. The global feature vector G encapsulates the discriminative information necessary for PReID tasks. This vector is passed through a fully connected layer to predict identity classifications or generate ranking scores. The hierarchical refinement strategy, combined with SE blocks, enables the network to extract robust, multi-resolution features for accurate PReID performance.

3.2 Swin transformer (ST) block

The ST block improves computational performance in vision problems by leveraging a shifted window-based self-attention mechanism. Instead of global attention, self-attention is computed within local, non-overlapping windows, tremendously reducing computational complexity. The computational cost for standard global self-attention is expressed as

$$\Omega(\text{MSA}) = 4h_{\text{img}}w_{\text{img}}C_{\text{chan}}^2 + 2(h_{\text{img}}w_{\text{img}})^2C_{\text{chan}}, \quad (2)$$

while the window-based self-attention reduces it to

$$\Omega(\text{W-MSA}) = 4h_{\text{img}}w_{\text{img}}C_{\text{chan}}^2 + 2M_{\text{win}}^2h_{\text{img}}w_{\text{img}}C_{\text{chan}}. \quad (3)$$

Here, $h_{\text{img}} \times w_{\text{img}}$ represents the image dimensions in patches, C_{chan} is the number of channels, and M_{win} (default $M_{\text{win}} = 7$) is the window size.

The shifted window partitioning further Optimizes the model by enabling cross-window interactions, improving the capture of long-range dependencies. Instead of attending to the entire image at once, the shifted window attention mechanism divides the image into smaller regions (windows) and shifts them slightly in each layer. This allows the model to capture both local and global features efficiently, reducing computational cost while maintaining context. This approach is particularly effective in handling occlusions and variations in appearance, as the model can dynamically focus on smaller, more relevant regions of the image. Additional components include Layer Normalization (LN) for input standardization, a two-layer Multi-Layer Perceptron (MLP) with GELU activation for feature refinement, and residual connections after the self-attention and MLP layers to ensure efficient gradient flow during training.

The computational complexity equations are:

$$\Omega(\text{MSA}) = 4h_{\text{img}}w_{\text{img}}C_{\text{chan}}^2 + 2(h_{\text{img}}w_{\text{img}})^2C_{\text{chan}}, \quad (4)$$

$$\Omega(\text{W-MSA}) = 4h_{\text{img}}w_{\text{img}}C_{\text{chan}}^2 + 2M_{\text{win}}^2h_{\text{img}}w_{\text{img}}C_{\text{chan}}. \quad (5)$$

Fig. 3 for an illustration of the Swin Transformer block, which highlights the shifted window-based self-attention mechanism and other key components.

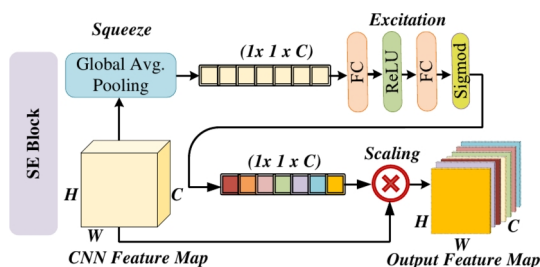


Fig. 4 Squeeze-and-Excitation (SE) block

3.3 SE-block channel attention

The SE mechanism boosts feature representation by assigning importance weights to each channel, thereby recalibrating channel-wise responses. This enables the model to suppress less essential features while emphasizing the most informative ones, improving its ability to capture discriminative features. Figure 4 illustrates the SE block's workflow, from input feature extraction through squeezing, excitation, and recalibration. The SE block operates in two key stages: *squeezing* and *excitation*.

3.4 Squeeze operation

In the first step, the input feature map's spatial dimensions are compressed to form a global channel descriptor. This is achieved using Global Average Pooling (GAP), which computes a channel-wise descriptor z_c for each channel c . Given a feature map $F \in \mathbb{R}^{H \times W \times C}$ with height H , width W , and C channels, GAP extracts the representative feature for each channel:

$$z_c = \text{GlobalAvgPool}(F) = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W F_{i,j,c} \quad (6)$$

The resulting vector $z \in \mathbb{R}^C$ contains the global average pooled values for each channel, representing the "squeezed" global information for each channel.

3.5 Excitation operation

The second step learns the channel-wise attention weights using a lightweight fully connected (FC) network, consisting of two layers. The input to the excitation operation is the squeezed vector z , which is first passed through a hidden layer with a ReLU activation function and then through a final sigmoid activation to produce the attention weights s_c for each channel. The excitation operation is expressed as:

$$z = \sigma(W_2 \cdot \text{ReLU}(W_1 \cdot z)) \quad (7)$$

where W_1 and W_2 are the weight matrices of the two FC layers, σ is the sigmoid activation function, and z is the final channel attention vector, where each element $z_c \in \mathbb{R}^C$ represents the attention weight for the corresponding channel.

3.6 Recalibration

The final step recalibrates the input feature map F by multiplying each channel by its corresponding attention weight. This process refines the most important channels while



Fig. 5 An object-aware random erasing, replacing patches with random pixel values

Require: Input image $I \in \mathbb{R}^{H \times W \times 3}$, number of stages N , number of patches P , learning rate `--lr`, batch size `--batchsize`, random erasing probability `--erasing-p`, warm-up epochs `--warm-epoch`.

Ensure: Identity prediction or ranking score.

- 1: **Initialization:**
- 2: Initialize Swin Transformer blocks $ST_1, ST_2, \dots, ST_N$, and Squeeze-and-Excitation blocks $SE_1, SE_2, \dots, SE_N$.
- 3: Initialize patch partition of I into $P_i \in \mathbb{R}^{4 \times 4 \times 3}$.
- 4: Initialize the feature map F_0 after embedding patches.
- 5: **for** each stage $i = 1$ to N **do**
- 6: **Step 1: Patch Partitioning:**
- 7: Divide I into non-overlapping patches P_i of size $4 \times 4 \times 3$.
- 8: **Step 2: Swin Transformer Block:**
- 9: **for** each block ST_i **do**
- 10: Compute self-attention for each patch using the formula:

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V$$
- 11: **end for**
- 12: **Step 3: SE Block:**
- 13: Apply SE block on output from ST_i :

$$F_{SE_i} = F_i \cdot \sigma(W_2 \cdot \text{ReLU}(W_1 \cdot \text{GAP}(F)))$$
- 14: **Step 4: Feature Extraction:**
- 15: Apply GAP to the final feature map F_N to obtain a compact feature vector $G \in \mathbb{R}^C$.
- 16: **Step 5: Identity Prediction:**
- 17: Pass G through a fully connected layer to get the identity prediction or ranking score.
- 18: **end for**
- 19: **Output:** Return identity prediction or ranking score based on G .

Algorithm 1 SwinReID.

suppressing the less relevant ones. The recalibrated feature map F_{SE} is obtained as follows:

$$F_{SE} = F \cdot z \quad (8)$$

Here, $F_{SE} \in \mathbb{R}^{H \times W \times C}$ represents the feature map with recalibrated channel-wise importance.

3.7 Optimization parameters & losses

We have applied six key techniques to improve generalization and robustness: RE, LW-up, CL, ArcFace Loss (AFL), CosFace Loss (CFL), and Contrastive Loss (CTL). These methods optimize the performance of the PReID network.

RE: The RE improves robustness by randomly occluding image patches, forcing the network to focus on diverse features. The modified image I' is:

$$I' = \begin{cases} I_{\text{erase}} & \text{with probability } p, \\ I & \text{with probability } 1 - p. \end{cases} \quad (9)$$

Object-aware RE targets both object and background, as shown in Fig. 5.

LW-up: The LW-up gradually increases the learning rate ($\alpha(t)$) during initial training to prevent abrupt weight updates:

$$\alpha(t) = \alpha_{\text{initial}} + \frac{(\alpha_{\text{target}} - \alpha_{\text{initial}})}{T} \cdot t, \quad (10)$$

where α_{initial} and α_{target} are the initial and target rates, T is the number of warm-up epochs, and t is the current epoch.

CL: The CL emphasizes harder positive/negative pairs to strengthen feature discriminability:

$$L = \frac{1}{N} \sum_{i=1}^N \sum_{j=1}^N [y_{ij} \max(0, m_1 + \Delta_i - \cos \theta_{ij}) + (1 - y_{ij}) \max(0, \cos \theta_{ij} - m_2)], \quad (11)$$

where $\cos(\theta_{ij})$ is cosine similarity, y_{ij} is the pair label, and m_1, m_2 are margins.

AFL: The AFL incorporates an angular margin for better inter-class separation.

$$L_{\text{AFL}} = -\log \frac{e^{s \cdot \cos(\theta_{y_i+m})}}{e^{s \cdot \cos(\theta_{y_i+m})} + \sum_{j \neq y_i} e^{s \cdot \cos(\theta_j)}}. \quad (12)$$

CFL: The CFL also modifies the target angle's cosine to magnify separation:

$$L_{\text{CFL}} = -\log \frac{e^{s \cdot (\cos(\theta_{y_i}) - m)}}{e^{s \cdot (\cos(\theta_{y_i}) - m)} + \sum_{j \neq y_i} e^{s \cdot \cos(\theta_j)}}. \quad (13)$$

CTL: The CTL pulls positive pairs together and pushes negative pairs apart:

$$L_{\text{CTL}} = \frac{1}{2N} \sum_{i=1}^N [y_{ij} \cdot d_{ij}^2 + (1 - y_{ij}) \cdot \max(0, m - d_{ij})^2], \quad (14)$$

where d_{ij} is the distance, y_{ij} is the label, and m is the margin. These techniques collectively maximize the network's performance in challenging re-identification tasks.

4 Experimental settings

This section presents the experimental settings for evaluating SwinReID. Section 4.1 introduces the datasets used, detailing their characteristics and relevance to the PReID task. Section 4.2 outlines the implementation details, including training configurations, hyperparameters, and optimization strategies. Section 4.3 evaluates SwinReID's performance and compares it with state-of-the-art (SOTA) methods using standard ReID metrics like CMC and mAP. Section 4.4 conducts ablation studies to assess the impact of key components, such as the SE-Block module, tuning parameters, and loss functions, to optimize the model's performance for PReID. Finally, Section 4.5 presents a case study demonstrating the practical application of SwinReID.

4.1 Datasets

This section summarizes the datasets used, including Market-1501, DukeMTMC-reID, MSMT17, and Occluded-Duke, as well as the experimental setup and configurations used to assess the proposed approach-Table 1 offers a statistical summary. We evaluate the proposed **SwinReID** network using four comprehensive PReID datasets: **(1) Market-1501** [39]. These datasets primarily focuses on general PReID in outdoor environments with multiple camera views. It consists of six cameras capturing 32,668 images of 1,501 pedestrians. The test set contains 19,732 images of 750 individuals, while the training set includes 12,936 images of 751 individuals. The query set is complete with 3,368 hand-drawn images. **(2) DukeMTMC-ReID**

[40]: This datasets introduces additional complexity with distractor individuals who appear in only one camera view, focusing on PReID tasks in a multi-camera setup. It contains 36,411 images from eight cameras, including 408 distractors and 1,404 individuals. The test set includes 17,661 images of 702 individuals across various camera views and 408 distractors, while the training set contains 16,522 photos of 702 individuals. The query set consists of 2,228 images of individuals who appear more than once. **(3) MSMT17** [41] This datasets is designed to reflect real-world challenges, such as varying pedestrian poses, illumination conditions, and backgrounds, captured across 15 cameras. It contains 126,441 images of 4,101 pedestrians, with 32,621 training images and 93,820 test images. The query set consists of 11,659 images, with the remaining forming the gallery. **(4) Occluded-Duke** [27] it focused on occluded PReID, this datasets includes 15,618 training images of 702 individuals, 17,661 gallery images, and 2,210 occluded query images from 1,110 pedestrians.

4.2 Implementation details

The ST, specifically the Swin base model with a patch size of 4 and a window size of 7, was used as the backbone in our final configuration for PReID. The model was initialized with pre-trained weights from ImageNet-22K, and the classification head was removed to focus on feature extraction. Additionally, a channel attention block (SE Block) was integrated to enhance the learning of discriminative features.

We experimented with different loss functions, such as CL, AFL, CFL, and CTL. After extensive experiments and fine-tuning with various settings and losses, the CL was selected as our final loss function for the proposed model. The training configurations included a batch size of 32, a learning rate 0.01, and a weight decay 0.0005. Stochastic Gradient Descent (SGD) with momentum was used as the optimizer, and a learning rate warm-up was applied during the first 3 epochs. Dropout was set to 0.5, and random erasing augmentation with a probability of 0.3 was used to prevent overfitting. Images were kept at their original resolutions to address varying image sizes across the datasets.

The proposed SwinReID network was trained on an NVIDIA Tesla V100 GPU using the PyTorch framework across 100 epochs. To ensure stable training, a LW-up was applied for the first three epochs. Data augmentation methods, such as 90-degree rotations and random horizontal flipping, were used to increase robustness. The CL function improved identification discrimination by optimizing the angular distances between feature vectors. The CL function is defined as:

Table 1 Summary of datasets including the number of identities, training, query, and gallery images

Datasets	ID	Train	Query	Gallery
Market-1501	1501	12,936	3368	16,364
DukeMTMC-reID	1404	16,522	2228	17,661
MSMT17	4101	32,621	11,659	82,161
Occluded-Duke	1404	15,618	2210	17,661

Table 2 Market-1501 and DukeMTMC-reID datasets

Backbone	Method	Market-1501		DukeMTMC-reID	
		R1	mAP	R1	mAP
CNN	DGNet [42]	94.8	86.0	86.6	74.8
	IANet [43]	94.4	83.1	87.1	73.4
	HPM [44]	94.2	82.7	86.6	74.3
	SAN [25]	96.1	88.0	87.0	75.7
	PCB [16]	93.8	81.6	83.3	69.2
	CI-CNN [45]	95.2	89.5	-	-
	Deep-Person [46]	92.3	79.6	80.9	64.8
	MLFN [47]	90.0	74.3	81.0	62.8
	PGFA [27]	91.2	76.8	82.6	65.5
	HOReID [48]	94.2	84.9	86.9	75.6
	AANet [49]	93.9	83.4	87.7	74.3
	CDNet [50]	95.1	86.0	88.6	76.8
Transformer	DeiT-B/16 [33]	94.4	86.6	89.3	78.9
	ViT-B/16 [33]	94.7	86.8	88.8	79.3
	Denseformer [51]	95.0	88.1	<u>90.0</u>	81.3
	DRL-Net [†] [52]	94.7	86.9	88.1	76.6
	FED [†] [53]	95.0	86.3	89.4	78.0
	CLIP-ReID [54]	95.4	<u>90.5</u>	90.8	<u>83.1</u>
	SwinReID	<u>95.5</u>	92.7	89.9	87.3

The best results are highlighted in bold, and the second-best results are underlined

$$L_{CL} = \frac{1}{N} \sum_{i=1}^N [\log(1 + \exp(\alpha(d_{\text{pos}} - d_{\text{neg}})))] \quad (15)$$

Where d_{pos} and d_{neg} are the distances between positive and negative pairs, respectively, and α is a scaling factor.

This configuration, optimized through experimentation, will be further discussed in the ablation study.

4.3 SwinReID evaluation & comparison with SOTA

This section evaluates SwinReID on four benchmarks: MSMT17, Market-1501, DukeMTMC-reID, and Occluded-Duke. Performance is measured using mAP and CMC top-K. A comparison with SOTA methods highlights SwinReID's superior mAP and rank-1 accuracy (R1), demonstrating its effectiveness in PReID. The best results are highlighted in bold, and the second-best results are underlined.

Market-1501 & DukeMTMC-reID On the Market-1501 datasets, SwinReID achieves a remarkable 92.7% mAP and 95.5% R1 accuracy, dramatically surpassing CNN-based models such as DGNet [42] (86.0% mAP, 94.8% R1) and SAN [25] (88.0% mAP, 96.1% R1). Compared to transformer-based models, SwinReID outperforms DeiT-B/16 [33] (86.6% mAP, 94.4% R1) and ViT-B/16 [33] (86.8% mAP, 94.7% R1) with an mAP improvement of 6.1% and 5.9%, respectively. These results highlight SwinReID's boosted ability to capture long-range dependencies

Table 3 Occluded-Duke datasets

Backbone & Method	R1	mAP
CNN		
PGFA [27]	51.4	37.3
HOReID [48]	55.1	43.8
ISP [55]	62.8	52.3
Ad-Occluded [56]	44.5	32.2
PVPM [24]	47.0	37.7
Transformer		
DeiT-B/16 [33]	60.6	53.1
ViT-B/16 [33]	60.5	53.1
Denseformer [51]	63.8	55.6
PAT [57]	<u>64.5</u>	53.6
CLIP-ReID [54]	67.2	60.3
SwinReID	64.1	64.3

The best results are highlighted in bold, and the second-best results are underlined

and optimally handle Occlusions and pose variations (see Table 2).

On the DukeMTMC-reID datasets, SwinReID achieves 87.3% mAP and 89.9% R1 accuracy, outperforming transformer-based networks like CLIP-ReID [54] (83.1% mAP, 90.8% R1) and Denseformer [51] (81.3% mAP, 90.0% R1). Specifically, SwinReID improves mAP by 4.2% over CLIP-ReID and by 6.0% over Denseformer, demonstrating its robustness across different re-identification tasks (see Table 2).

Occluded-Duke SwinReID also performs well on the Occluded-Duke datasets, achieving 64.3% mAP and 64.1% R1 accuracy. While transformer-based methods such as CLIP-ReID [54] (60.3% mAP, 67.2% R1) demonstrate promising results, SwinReID outperforms them by a 4.0% increment in mAP. Additionally, CNN-based methods such as ISP [55] (52.3% mAP, 62.8% R1) and PGFA [27] (37.7% mAP, 51.4% R1) fall short in comparison, showcasing SwinReID's resilience to Occlusions and partial visibility challenges (see Table 3).

MSMT17 SwinReID achieves 87.5% mAP and 91.63% R1 on the challenging MSMT17 datasets, which contains notable variations in viewpoint, illumination, and Occlusion. The proposed SwinReID surpasses CLIP-ReID [54] (75.8% mAP, 89.7% R1) and Denseformer [51] (65.3% mAP, 84.1% R1), highlighting its capacity to manage complex and diverse re-identification scenarios, such as drastic appearance changes due to lighting or pose variations. All results are shown in Table 4.

Our proposed SwinReID consistently outperforms existing methods in both mAP and R1 accuracy across all four datasets, setting a new benchmark in PReID. Its superior performance in handling pose, Occlusion, and illumination variations, combined with advanced feature extraction and attention mechanisms, positions it as a leading solution.

Table 4 MSMT17 datasets

Backbone & Method	R1	mAP
CNN		
DGNet [42]	77.2	52.3
IANet [43]	75.5	46.8
SAN [25]	79.2	55.7
PCB [16]	68.2	40.4
CDNet [50]	78.9	54.7
Transformer		
DeiT-B/16 [33]	81.9	61.4
ViT-B/16 [33]	81.8	61.0
Denseformer [51]	84.1	65.3
CLIP-ReID [54]	<u>89.7</u>	<u>75.8</u>
SwinReID	91.63	87.5

The best results are highlighted in bold, and the second-best results are underlined

Table 5 mAP scores and Rank-1 accuracy for the proposed SwinReID model

Datasets	mAP Score	Rank-1 (%)
DukeMTMC-reID	87.3	89.9
Market-1501	92.7	95.5
MSMT17	87.5	91.63
Occluded-Duke	64.1	64.3

The best results are highlighted in bold, and the second-best results are underlined

SwinReID excels over transformer-based models like CLIP-ReID, Denseformer, and CNN-based methods, demonstrating its effectiveness in both conventional and occluded re-identification tasks. These results highlight SwinReID as a reliable and efficient choice for real-world applications, balancing high accuracy with computational efficiency, as summarized in Table 5.

4.4 SwinReID: performance and challenges

SwinReID outperforms CNN-based methods on occluded datasets due to its shifted window attention mechanism, which captures both local and global features, enabling robust handling of occlusions. Unlike CNNs, which rely on fixed receptive fields and local feature extraction, SwinReID dynamically shifts its attention to relevant regions, preserving contextual understanding under occlusion. CNNs struggle with long-range dependencies, limiting their performance in complex ReID scenarios involving pose changes or partial occlusions. The SE-block further amplifies SwinReID by adaptively recalibrating channel-wise

feature importance, suppressing irrelevant information, and emphasizing discriminative features. This improves intra-class compactness and inter-class separability, leading to better performance.

However, despite these strengths, SwinReID may fail in cases of extreme occlusion or when two individuals wear nearly identical clothing, particularly in scenarios where clothing and background are highly similar. These failure cases highlight the challenges of ReID in real-world scenarios, where fine-grained distinctions are critical for accurate identification.

4.5 Evaluation of parametric variants of SwinReID

In this section, we provide a comprehensive evaluation of our proposed SwinReID architecture across two parametric variants: Small (SwinReID_S) and Large (SwinReID_L) on the Market-1501 dataset, using an NVIDIA Tesla V100 GPU. Among these, our proposed **SwinReID** model is specifically optimized to balance computational efficiency with high performance.

To assess the effectiveness of each model, we compare their performance across several key metrics: total parameters(T-Params), multiply-accumulate operations (MACs), inference time (iTime), GPU memory consumption, R1 accuracy, and mean Average Precision (mAP). Computational costs were measured using the OpCounter toolkit.

Our proposed SwinReID model demonstrates the best balance between computational efficiency and recognition accuracy, making it the optimal choice for real-world PReID tasks. With a R1 accuracy of 95.57% and an mAP of 92.80%, it considerably outperforms the lightweight SwinReID_S, which suffers from a 4.01% drop in R1 accuracy and a 5.01% lower mAP, despite its lower computational cost.

While SwinReID_L offers comparable accuracy, it comes at a high computational cost-requiring 57.6% more GPU memory and 53.2% longer inference time than our proposed model. This makes SwinReID_L less suitable for real-time applications. In contrast, our proposed SwinReID model achieves high recognition performance while optimizing computational demands, making it the most efficient and effective solution in terms of both accuracy and resource utilization. Table 6 presents a comparative analysis of the SwinReID variants.

Table 6 Evaluation of parametric variants of SwinReID: Small, Large, and proposed SwinReID

Model	T-Params (M)	MACs (G)	iTime (s)	GPU (MB)	mAP	R1
SwinReID_S	28.37	4.37	2.08	3,880.02	87.79%	91.86%
SwinReID_L	196.46	34.08	6.97	13,557.97	90.51%	93.21%
SwinReID	87.79	15.17	4.55	8,598.35	92.80%	95.57%

The best results are highlighted in bold, and the second-best results are underlined

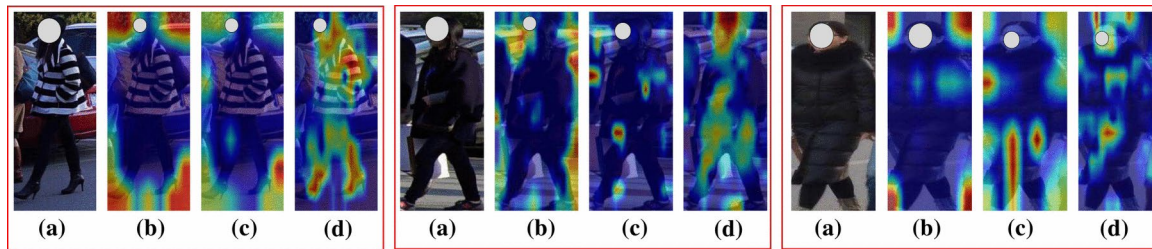


Fig. 6 Grad-CAM heatmap visualization results of SwinReID variants: Small, Large, and proposed SwinReID. **a** Original images; **b** Small SwinReID results; **c** Large SwinReID results; and **d** Proposed SwinReID results

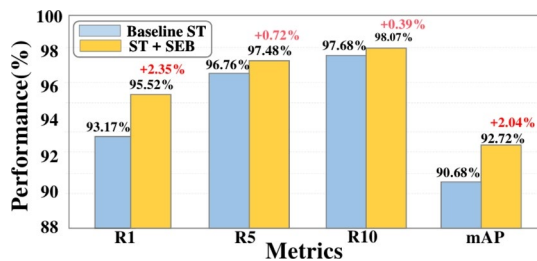


Fig. 7 Performance metrics with and without SE-Block integration

4.6 Heatmap visualization

The heatmap visualization in Fig. 6 illustrates the Grad-CAM visualization results of SwinReID_S, SwinReID_L, and the proposed SwinReID model. The SwinReID_S focuses on localized regions but often includes background noise, while the SwinReID_L captures more global features but struggles to identify discriminative regions, often emphasizing irrelevant areas precisely.

In contrast, the proposed SwinReID focuses more on the main parts of the human body (such as the head, chest, arms, legs, feet, etc), adeptly capturing discriminative features while minimizing background interference. This ability to balance global and local context enables the proposed model to achieve superior performance in PReID tasks.

4.7 Ablation work

This section presents extensive ablation work to substantiate the novelty and significance of the introduced SwinReID model for PReID. The investigation focuses on the integration of the channel attention mechanism, leveraging the SE-Block technique, and the fine-tuning of hyperparameters, including RE, LW-up, and alternative learning strategies. Our approach achieves substantial improvements by embedding these enhancements within the ST backbone and introducing the novel channel attention mechanism. Experimental results in the Market datasets underscore the effectiveness of the model, consistently outperforming their SOTA counterparts. The highest-performing metrics are emphasized in bold for clarity.

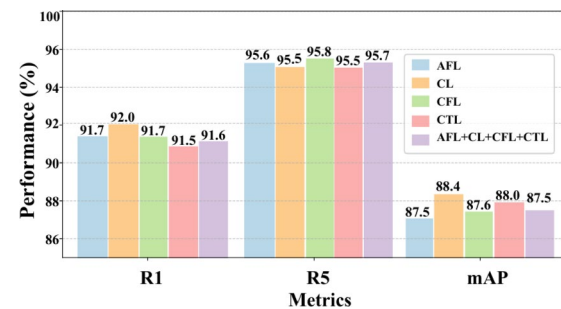


Fig. 8 Circle loss vs other loss functions comparison

4.8 Impact of SE-Block module

The integration of the SE-Block channel attention mechanism considerably boosted the performance of our PReID system, yielding notable gains in both accuracy and mAP. The results can be seen in Fig. 7. The inclusion of the SE-Block (SEB) within the backbone ST resulted in a 2.35% increase in Rank-1 accuracy, a 0.72% increase in Rank-5 accuracy, and a 0.39% increase in Rank-10 accuracy. Furthermore, the mAP improved by 2.04%, highlighting the SEB's effectiveness in enhancing the model's ability to capture discriminative features. These results demonstrate that the SEB substantially optimizes the backbone ST by refining channel-wise attention, which is crucial for robust feature extraction in the PReID task.

4.9 Tuning parameters & losses

To optimize the performance of our proposed SwinReID model for PReID, we conducted an ablation study to fine-tune key hyperparameters, such as batch size, learning rate, RE probability, and LW-up period. The optimal configuration found was a batch size of 32, a learning rate of 0.001, a RE probability of 0.3, and a warm-up period of 3 epochs.

Among the loss functions tested, CL performed the best on the Market datasets, achieving 92.0% for R1 (Top-1 accuracy) and 88.4% for mAP, as reported at the 80th epoch (see Fig. 8). CL outperformed AF by +0.4% in R1 and +1.0% in mAP. CL's advantage lies in its dynamic adjustment of positive and negative pairs based on similarity scores,

prioritizing hard samples for balanced optimization. Unlike fixed-margin losses like AFL and CFL, as well as pairwise methods such as CTL, CL enriches intra-class compactness and inter-class separability, addressing PReID challenges, including occlusions and pose variations.

Furthermore, we examined the effect of increasing the learning rate from 0.001 to 0.01 while keeping other settings constant. As shown in Table 7, the learning rate of 0.01 remarkably refined performance, with R1 rising to 95.5, R5 to 97.5, and mAP to 92.7. These results indicate that the higher learning rate facilitated better convergence and generalization.

Given these findings, we chose a learning rate of 0.01 for training, as it consistently yielded the best results across all evaluation metrics. A comparison of CL at learning rates 0.001 and 0.01 is presented in Table 8.

4.10 Case study

This section evaluates SwinReID's performance across training epochs and challenging scenarios. **Case 1** analyzes accuracy improvements over epochs, showing amplified feature extraction and outperforming baseline. **Case 2** assesses robustness under occlusions, illumination changes, and viewpoint variations, demonstrating high retrieval accuracy in all cases.

4.11 Case 1: Evaluation by epochs

The performance of the proposed network was evaluated across different training epochs, with results presented in Table 9. Compared to the baseline ST, the proposed network showed remarkable improvements in Rank-1 accuracy and mean average precision (mAP).

At the 80th epoch, the proposed network achieved a Rank-1 accuracy of 95.25% and an mAP of 92.54%, outperforming the baseline by 2.11% and 1.89%. The 90th epoch saw the best performance with a Rank-1 accuracy of 95.22% and mAP of 92.64%, improving by 2.00% and 1.89% over the baseline ST. At the 100th epoch, the network maintained high performance, achieving 95.52% Rank-1 accuracy and 92.72% mAP, surpassing the baseline by 2.36% and 1.04%. These results validate the proposed modifications' effectiveness in enhancing feature extraction and overall performance.

4.12 Case 2: evaluation by queries

This section evaluates the performance of SwinReID under three challenging scenarios: partial occlusions, illumination variations, and viewpoint changes. The results demonstrate

Table 7 Circle Loss (CL) Performance with learning rate 0.01

Epochs	R1	R5	mAP
50	94.63%	97.06%	91.86%
60	94.60%	97.12%	91.89%
70	<u>95.31%</u>	97.39%	92.43%
80	95.25%	97.48%	92.54%
90	95.22%	97.62%	<u>92.64%</u>
100	95.52%	97.48%	92.72%

The best results are highlighted in bold, and the second-best results are underlined

Table 8 Comparison of Circle Loss at learning rates 0.001 vs. 0.01

Learning Rate	R1	R5	mAP
0.001	92.0	95.5	88.4
0.01	95.5	97.5	92.7

The best results are highlighted in bold, and the second-best results are underlined

Table 9 Comparison of the baseline ST and our proposed SwinReID, with the best performance highlighted

Model	Epoch	R1	R5	mAP
ST	50	92.0%	96.3%	88.4%
	60	91.6%	95.1%	87.0%
	70	93.0%	96.1%	89.9%
	80	93.1%	<u>96.6%</u>	90.5%
	90	93.7%	96.8%	90.8%
	100	<u>93.2%</u>	96.8%	<u>90.7%</u>
Our	50	94.6%	97.1%	91.9%
	60	94.6%	97.1%	91.9%
	70	95.3%	97.4%	92.4%
	80	<u>95.3%</u>	97.5%	92.5%
	90	95.2%	97.6%	<u>92.6%</u>
	100	95.5%	<u>97.5%</u>	92.7%

The best results are highlighted in bold, and the second-best results are underlined

SwinReID's robustness, as it achieves high retrieval accuracy in all of these cases.

Partial Occlusions: As shown in Fig. 9a, SwinReID correctly retrieved all matching identities, even in the presence of severe occlusions.

Illumination Variations: As shown in Fig. 9b, SwinReID successfully handles illumination variations, successfully retrieving relevant individuals even under challenging lighting conditions. For example, the query "0007_c5s3_031580_0" yielded 8 matches, demonstrating the model's robustness to changes in lighting.

Viewpoint Changes: SwinReID also exhibits strong performance under extreme viewpoint changes, as shown in Fig. 9c. For instance, the query "0021_c1s6_032261_0" retrieved nine matches, highlighting the model's ability to handle significant changes in perspective.

These results highlight SwinReID's strengths in real-world scenarios while revealing limitations in challenging conditions, such as extreme illumination, viewpoint



Fig. 9 Market-1501 retrieval results for challenging queries. Green titles indicate correct matches with SwinReID

variations, occlusions, and cases involving partial body parts or similar appearances, as shown in Fig. 9.

5 Conclusion

This paper proposed SwinReID, a novel person-reidentification model combining the ST with a lightweight channel attention mechanism via the SE block. The SE block refines feature extraction by recalibrating channel-wise attention, addressing the limitations of traditional CNNs in capturing long-range dependencies. Extensive ablation studies validated the contributions of both the SE block and Swin Transformer. SwinReID achieved state-of-the-art results on benchmark datasets, including 95.5% rank-1 accuracy and 92.7% mAP on Market-1501, and 89.9% rank-1 accuracy and 87.3% mAP on DukeMTMC-reID. Future work will focus on advanced attention mechanisms, self-supervised learning, and real-time optimization for edge computing environments.

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Author contributions TAA: conceptualization, methodology, and writing-analysis. ZP & MQ: conceptualization and supervision, and AUM reviewed the manuscript.

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Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships appeared to influence the work reported.

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